

PENTELEU, Romania

WMO: 152840

Lat: 45.6028N

Lon: 26.4097E

Elev: 5359

StdP: 12.06

Time Zone: 2.00 (EUE)

Period: 88-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.6	(c) 3.7	(d) -13.0	(e) 3.3	(f) 18.4	(g) -5.9	(h) 4.9	(i) 16.3	(j) 61.9	(k) 10.4	(l) 53.1	(m) 15.6	(n) 20.2	(o) 320	(p) 0.840

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.8	(c) 69.2	(d) 58.5	(e) 66.6	(f) 57.0	(g) 64.3	(h) 55.8	(i) 60.7	(j) 66.7	(k) 59.0	(l) 64.6	(m) 57.4	(n) 62.7	(o) 10.0	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.3	88.9	63.5	56.7	83.8	61.7	55.3	79.4	59.8	29.6	66.7	28.3	64.6	27.1	62.8	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 41.0	(b) 34.6	(c) 30.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	19.8	23.1	26.3	34.7	45.1	52.6	54.7	55.8	46.8	39.6	31.9	23.5
	DBStd	15.08	9.24	8.99	8.49	10.09	6.86	6.59	5.83	6.57	8.56	9.39	9.30	9.12
	HDD50	5039	936	753	734	466	179	48	22	27	161	346	545	822
	HDD65	9897	1401	1173	1199	909	616	374	323	288	546	788	993	1287
	CDD50	626	0	0	0	7	28	126	167	207	66	23	2	0
	CDD65	8	0	0	0	0	0	2	2	3	1	0	0	0
	CDH74	15	0	0	0	0	0	2	10	2	1	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	13.8	16.7	15.8	15.0	13.8	11.9	11.9	11.6	9.9	12.7	13.4	15.6	17.2
Precipitation	PrecAvg	27.6	1.3	1.1	1.5	2.2	3.3	4.1	4.0	3.3	2.2	1.9	1.4	1.4
	PrecMax	37.8	2.8	2.9	2.8	4.4	7.0	7.4	8.9	6.9	4.8	5.2	3.2	2.8
	PrecMin	16.5	0.4	0.1	0.2	0.9	1.5	1.7	1.5	0.9	0.5	0.1	0.0	0.3
	PrecStd	5.4	0.7	0.6	0.6	1.1	1.6	1.6	1.9	1.5	1.1	1.2	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	46.5	50.0	60.0	64.5	71.9	73.4	72.0	70.1	63.2	54.8	48.8
		MCWB	33.5	39.1	39.7	46.7	52.3	59.9	60.3	60.1	57.5	47.9	42.9	37.1
	2%	DB	36.8	41.2	45.6	55.2	60.6	66.9	68.2	68.9	64.1	59.0	50.6	42.4
		MCWB	32.9	33.6	37.8	45.5	50.6	57.3	59.1	58.3	54.4	47.5	41.8	35.7
	5%	DB	34.1	37.6	41.6	51.5	57.3	63.9	65.7	66.3	60.7	55.5	47.3	38.5
		MCWB	31.3	33.8	35.9	43.8	49.7	56.6	57.4	56.5	52.6	45.5	40.1	34.8
	10%	DB	31.7	34.6	37.7	47.9	54.3	61.3	63.0	64.0	57.4	52.1	44.4	35.3
		MCWB	29.6	31.2	33.8	41.0	47.7	54.9	56.3	55.5	50.8	44.2	38.8	32.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	41.7	41.1	49.3	54.6	62.1	62.1	62.4	59.4	54.3	46.2	39.9
		MCDB	37.6	46.1	48.0	58.2	59.7	68.8	68.9	69.8	67.8	59.7	50.9	43.1
	2%	WB	34.3	36.7	38.8	46.9	52.5	59.7	60.3	60.3	55.9	50.2	44.3	37.4
		MCDB	36.4	40.2	44.4	53.5	57.4	64.9	66.3	66.7	62.5	55.3	48.1	39.8
	5%	WB	32.1	34.4	37.0	45.1	50.6	57.7	58.9	58.3	53.7	47.7	42.4	35.3
		MCDB	33.3	37.3	41.7	50.9	55.5	62.7	63.7	64.4	59.6	53.9	46.0	38.4
	10%	WB	29.9	32.2	34.7	42.8	48.8	56.1	57.2	56.9	52.1	45.7	40.2	32.8
		MCDB	31.3	33.9	38.1	47.9	53.5	60.8	62.1	62.5	57.4	50.7	43.1	35.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.8	10.4	11.0	11.6	11.1	11.1	10.8	9.7	9.5	8.8	8.8
		MCDBR	10.3	11.7	14.6	14.9	15.1	12.9	13.7	12.3	12.8	13.2	11.4	12.0
		MCWBR	7.7	8.0	9.2	9.7	8.5	8.1	8.4	7.8	8.3	10.5	7.6	7.0
	5% WB	MCDBR	9.3	10.4	13.1	14.0	12.9	12.4	12.5	11.4	11.4	12.5	9.8	9.4
		MCWBR	8.0	7.7	8.7	9.4	8.3	8.2	8.9	8.3	8.0	10.3	7.6	7.0
Clear-Sky Solar Irradiance	taub		0.258	0.280	0.311	0.356	0.340	0.341	0.352	0.361	0.330	0.303	0.283	0.259
	taud		2.439	2.391	2.396	2.308	2.397	2.406	2.371	2.342	2.419	2.496	2.515	2.466
	Ebn at Noon		276	288	293	286	294	293	288	281	282	276	264	263
	Edh at Noon		22	28	32	38	36	36	37	37	31	25	20	19
All-Sky Solar Radiation	RadAvg		413	672	1035	1407	1768	1876	1910	1728	1254	820	451	345
	RadStd		49	79	108	144	148	147	146	123	115	94	43	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	+1.06	N/A	N/A	+1.19	N/A	N/A	-229	-306	+140	+49

Nomenclature: See separate page